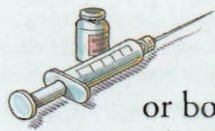


DRUGS



A DRUG IS ANY SUBSTANCE that, when put into the body, alters its normal workings or body chemistry. Natural body hormones, such as insulin, can act as drugs when taken in concentrated form. Medical drugs have many uses. Some, such as cough suppressants, may relieve symptoms; others, such as analgesics, deaden pain; while others, including antibiotics, treat the cause of disease. Drugs may also be taken for non-medical reasons, such as steroids to enhance sports performance and body-building. The abuse of such drugs may be illegal, and can cause physical harm.

Types of drugs

Drugs can be grouped by their medical uses or effects. For example, antibiotics kill bacteria, analgesics deaden pain, anti-inflammatories reduce swelling, anti-pyretics lower body temperature, and anti-coagulants help to prevent unwanted blood clots. Some drugs, such as aspirin, can be placed in more than one category.



Syringe containing cytotoxic drugs



Antibiotic

These drugs kill or disable germs (harmful microbes) known as bacteria. Most come from chemicals made either by fungi, or by other bacteria.

Antibiotic cream

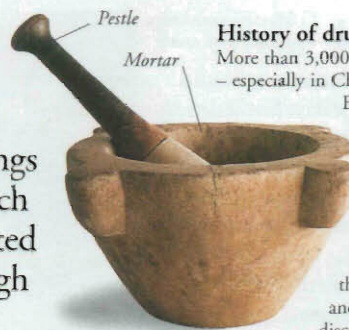
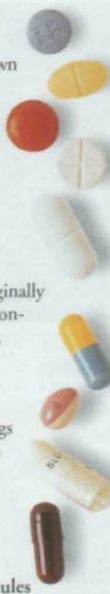
Analgesic

Painkillers come in two types: narcotics, such as morphine, codeine, and other opiates originally from the opium poppy; and non-narcotics, such as paracetamol, which have a different origin.

Cytotoxic

The name means "cell-poisoners", but cytotoxic drugs are designed to affect only the out-of-control cells in tumours and malignancies (cancers), while leaving normal body cells unharmed. They are one type of anti-cancer drug. They are very powerful and their doses and uses must be carefully supervised.

Tablets and capsules



History of drugs

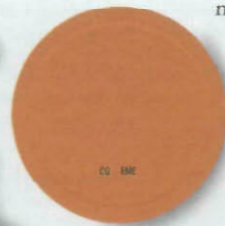
More than 3,000 years ago, people across the world – especially in China, India, the Middle East, Europe, and North Africa – used hundreds of different substances as drugs. They included herbal and mineral extracts, and animal products, such as blood, bile, and urine. Physicians mixed these drugs using a pestle and mortar, and often combined their use with magic, superstition, and religion. Modern research has discovered that some are effective.

How drugs work

Drugs change the processes within the cells of the body. Their effectiveness depends on the dose (quantity), and method of administration (or route into the body). These routes include: absorption through the skin from a cream or a skin patch; injections into a muscle, vein, or under the skin; inhalation; eye or ear drops; or the oral route, where medication is swallowed as tablets, pills, capsules, or liquid.



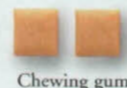
Pressurized inhaler



Transdermal patches



Syrup



Chewing gum



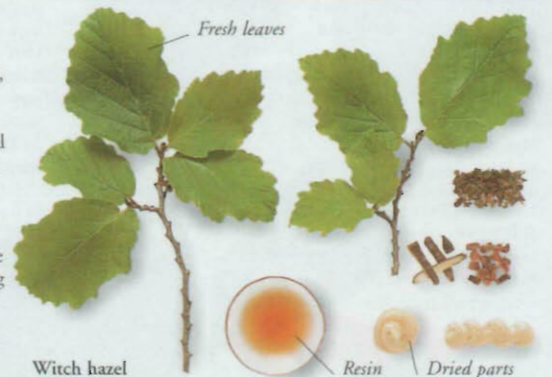
Suppositories



Pills and tablets

Drugs from nature

Half of modern drugs originate from plants, fungi, animals, or microbes. In ancient times, people were unable to separate the actual drug – the active ingredient – from its source. As chemistry became more sophisticated, scientists identified and purified these ingredients making the drug safer. Some drugs extracted originally from nature are now made from genetically engineered microbes.



Witch hazel

Resin

Dried parts

Paul Ehrlich

The German scientist Paul Ehrlich (1854–1915) dreamed of finding a substance that would act as a "magic bullet", by destroying invading germs, while leaving healthy body cells unaffected. He pioneered synthetic drugs (chemical agents made in the laboratory, rather than extracted from natural sources). The first of these was Salvarsan, which was a laboratory-made drug containing arsenic; it was effective against syphilis and related infections.



Drug research

In the laboratory, scientists analyse potential new drugs. They perform tests on the drug to establish its chemistry, and how it affects the body's processes. Then they test it on tissues and cells in the laboratory, on animals, and finally on human volunteers in clinical trials.

Drug research laboratory



Aspirin

Brand name – the name by which manufacturers sell a drug, e.g. Aspro.

Generic name – the name by which the active ingredient is known, e.g. aspirin.

Common chemical name – showing the chemical subgroups, e.g. acetyl-salicylic acid.

Chemical formula – lists the atoms and their numbers in the drug, e.g. $C_9H_8O_4$.